Modelling and Parameter Estimation of Diesel Engine Aftertreatment System under Catalyst Saturation for Aging Diagnostics

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This paper presents a model-based statistical framework for detecting catalyst aging in diesel engine SCR systems. A dynamic model for NO_x reduction under catalyst saturation with ammonia is derived from molar conservation, accounting for the interplay between residence time and sensor sampling frequency. The model parameters are estimated via maximum likelihood using the bounding relationship between actual and predicted NO_x reduction under saturation while handling uncertainties from NO_x sensor cross-sensitivity in the absence of ammonia sensors. The estimated parameters are shown to be sensitive to the catalyst's aging state, enabling a Wald test-based hypothesis-testing framework for aging detection. The proposed detector is validated using test-cell data at different aging levels and real-world data from four long-haul trucks.

Keywords: Diesel Engine Aftertreatment System, System Identification, Switched Nonlinear Systems, Parameter Estimation, Convex Optimization, Statistical Estimation and Detection.

1 Introduction

Diesel engine aftertreatment systems reduce the concentrations of harmful gases such as NO_x and CO from exhaust emissions. The Selective Catalytic Reduction (SCR) system reduces the engine-out NO_x to N_2 and H_2O using ammonia from urea dosing in the presence of a catalyst. This catalytic conversion process is regulated to decrease the levels of ammonia in the exhaust (ammonia slip) via two methods. The first is feedback control, which adjusts the urea injection rate based on the exhaust NO_x concentrations. The second method involves an additional catalytic reaction, the Ammonia Slip Catalyst (ASC), which is designed to oxidize any excess ammonia at the end of the SCR bed. Figure-1 shows a schematic of the SCR-ASC system. It has been found that the NO_x-reduction capacity of the catalyst decreases over time due to hydrothermal aging and deposition of contaminants such as excess urea, platinum, sulfur and phosphorus [1]. The aging of the catalyst leads to increased emissions of NO_x and ammonia slip, which can result in non-compliance with environmental regulations. A fault detection system for catalyst aging would provide better control over system maintenance and improve the overall reduction in emissions.

Fig. 1 Schematic of the SCR-ASC system

Thus, modern diesel aftertreatment systems, particularly those that integrate Selective Catalytic Reduction (SCR) with Ammonia Slip Catalyst (ASC), necessitate advanced onboard diagnostics (OBD) tools for accurate assessment of catalyst aging. However, the effectiveness of traditional OBD approaches for this purpose has been impeded by the absence of model validation with real-world catalyst degradation data, the limitations imposed by existing commercial NO_x sensors' cross-sensitivity to ammonia, and the absence of ammonia sensors in commercial vehicles.

Numerous studies have been conducted on modeling SCR-ASC systems and their control [2]. A prevalent modeling approach is to approximate the PDE model for the plug flow reaction as a set of ODEs using a sequence of Continuously Stirred Tank Reactors (CSTRs) ([3] , and [4]). The single CSTR approach was first justified in [5] and a nonlinear model was developed using these assumptions, which was then linearized for feedback control design ([6]). With this model, observers were designed to estimate the states corresponding to the catalyst's storage ([7], [8]). A method for detecting the catalyst's aging using an observer for the change in the maximum storage capacity of the catalyst, modeled as an exponential function of temperature, was also proposed in [7] and [9]. Alternatively, an onboard diagnostics (OBD) logic for catalyst aging based on the tailpipe NO_x and ammonia is proposed in [1] based on the hypothesis of reduction in adsorption sites and demonstrated on real-world trucks. A common theme in these studies is that the system identification results in a non-convex, nonlinear parameter estimation problem. Moreover, these studies assume the availability of all the gaseous states at the tailpipe for state estimation and to eliminate the effects of cross-sensitivity of the NO_x sensors, which is not always the case in real-world applications as tailpipe ammonia measurements are not generally available in commercial vehicles. Another fundamental issue with the CSTR model in the context of sensor sampling is that the discretization of the continuous model does not account for the residence time of the reactants [10], resulting in high sampling frequency requirements that are not feasible in practice. Additionally, the discretization requires at least two CSTRs to capture the system dynamics and causality [11], thereby increasing the model order.

An alternative approach to the problem would be discarding the CSTR assumption and modelling the time evolution of the sensor signals when a "plug" or "parcel" of the exhaust gases flows through the chamber in discrete time, considering the interplay between sampling and residence time of the reactants. The catalyst saturation that is inherently considered in the CSTR approach needs to be explicitly included as a separate mode of the system.

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This work develops such a nonlinear model for NO_x reduction under catalyst saturation, which serves as one of the modes within a complete switched nonlinear framework. Following the model development, we present an algorithm to estimate the parameters of the saturated catalyst model using real-world data. This estimation explicitly accounts for unknown operating modes, uncertainties arising from NO_x sensor cross-sensitivity to ammonia, and the absence of dedicated ammonia sensors. Utilizing these estimates, we introduce a catalyst mode detection algorithm to characterize the statistical properties of the parameter estimates. The limitations of both the estimation method and this mode detection approach are discussed and demonstrated in the appendix. Finally, we develop a statistical test for detecting catalyst aging based on these estimated parameters, with its performance demonstrated using both test-cell and real-world truck data.

2 SCR-ASC System Dynamics Under Catalyst Saturation

The SCR-ASC subsystem in the diesel engine aftertreatment systems involves reacting flow over a catalyst that adsorbs ammonia from urea decomposition. The reaction kinetics start with the decomposition of urea into ammonia followed by its adsorption onto the surface of the catalyst. The NO_x entering the chamber is reduced by this adsorbed ammonia. The Eley-Rideal reaction mechanism [12], [2], [3], [4] is considered for modeling the SCR reaction kinetics, where one reactant is in the gas phase (NO_x), and the other is adsorbed on the catalyst surface (NH_3). Reference [4] lists all the reactions that take place inside the SCR-ASC chamber. Ammonia oxidation (AMOX) happens on the surfaces of both the SCR and the ASC catalysts, but the SCR catalyst favors NO_x reduction while the ASC is specifically designed for ammonia oxidation. In order to keep the model order reasonably low and parameters identifiable, only the following three prominent reactions [5],[3] are considered:

1. Standard SCR reaction:

$$4NH_3^{ads} + 4NO + O_2 \xrightarrow{k_{scr}} 4N_2 + 6H_2O \tag{1}$$

2. Ammonia Oxidation:

$$4NH_3^{ads} + 3O_2 \xrightarrow{k_{oxi}} 2N_2 + 6H_2O \tag{2}$$

3. Ammonia Adsorption/Desorption:

$$NH_3 + \Theta_{free} \underset{k_{des}}{\overset{k_{ads}}{\rightleftharpoons}} NH_3^{ads} \tag{3}$$

This aggregation of reactions results in errors in parameter estimates; specifically, parameters containing the rate constant for the NO_x reduction (k_{scr}) have higher (bounded) uncertainty because they become dependent on the nitrogen selectivity of the AMOX reaction [13]. Considering the reaction chamber as a control volume (Figure-2), the dynamic model is developed based on the molar conservation of species at the inlet and outlet separated by a residence-time interval. By assuming a zero-order hold on the inlet species over the sampling time, the equations relating inlet and outlet concentrations at the residence-time scale are extended to the sampling-time scale. The residence time can be parameterized as a function of mass flow rate (F) assuming that the effect of the density variation on residence time is insignificant:

Fig. 2 Control volume in plug flow reaction

$$\tau(k) = \frac{\rho V_{scr}}{F(k)} = \frac{\tau_0}{F(k)} \tag{4}$$

Let $\{\bullet\}$ be the number of moles and $[\bullet]$ be the concentration of a given species $\bullet$. We have the molar conservation of NO_x across the control volume (V_{scr}) within one residence time (τ):

$$\{NO_x\}^{out}(t+(i+1)\tau) = \{NO_x\}^{in}(t+i\tau) + V_{scr}\int_0^{\tau}\frac{d}{dt}[NO_x]^{scr}\,dt \tag{5}$$

The number of moles entering at the inlet and leaving at the outlet can be written as a function of the concentration, residence time, and mass flow rate (F):

$$\{NO_x\}^{in/out} = [NO_x]^{in/out}\,\tau\left(\frac{F}{\rho}\right) = [NO_x]^{in/out}\,V_{scr} \tag{6}$$

Using the Eley-Rideal mechanism for the standard SCR reaction (1):

$$\frac{d}{dt}[NO_x]^{scr}\,dt = -k_{scr}\,[NH_3]^{ads}\,[NO_x]^{in} \tag{7}$$

Rewriting (5) in terms of concentrations and integrating over the residence time:

$$[NO_x]^{out}(t+(i+1)\tau) = [NO_x]^{in}(t+i\tau) - \tau k_{scr}\,[NH_3]^{ads}\,[NO_x]^{in} \tag{8}$$

Summing the equations from (8) for $i = 0$ to $n-1$, where n is the total number of residence times within one sampling period ($n = t_s/\tau$) and using the zero-order hold assumptions at the inlet (9) and outlet (10) concentrations, we obtain (11).

$$[NO_x]^{in}(t+i\tau) = [NO_x]^{in}(t) \quad \forall i \in [0, n-1] \tag{9}$$

$$[NO_x]^{out}(t+(i+1)\tau) = [NO_x]^{out}(t+t_s) \quad \forall i \in [0, n-1] \tag{10}$$

Letting tailpipe NO_x concentration be $x_1(k+1) = [NO_x]^{out}(t+t_s)$ and inlet NO_x concentration be $u_1(k) = [NO_x]^{in}(t)$, we get,

$$\underbrace{[NO_x]^{out}(t+t_s)}_{x_1(k+1)} = \underbrace{[NO_x]^{in}(t)}_{u_1(k)} - \tau(t)k_{scr}(t)\,[NO_x]^{in}(t)\underbrace{\left\{\frac{1}{n}\sum_{i=0}^{n-1}[NH_3]^{ads}(t+i\tau)\right\}}_{\gamma(k)} \tag{11}$$

The quantity $\gamma(k)$ is the average concentration of ammonia adsorbed on the catalyst's surface across the n residence times between k and $k+1$ samples. $\gamma(k)$ is bounded by the total concentration of voids on the surface of the catalyst, Γ, which decreases with increasing temperature.

$$0 \le \gamma(k) \le \Gamma \qquad \forall k \tag{12}$$

The unbounded dynamics of γ, $\gamma^{ub}(k)$ are a nonlinear function dependent on the concentration of adsorbed ammonia, urea dosing (u_{inj}), and the inlet NO_x concentration from the previous time step [10].

$$\gamma^{ub}(k) = g_\gamma\left(\gamma(k-1), u_{inj}(k-1), [NO_x]^{in}(k-1)\right) \tag{13}$$

The actual expression, which is beyond the scope of this work, can be derived using a similar approach to that used to derive NO_x

process dynamics (11). $\gamma(k)$ can be described using the following piecewise function:

$$\gamma(k) = \begin{cases} \gamma^{ub}(k) & \text{if} \quad 0 \le \gamma^{ub}(k) \le \Gamma \\ \Gamma & \text{if} \quad \gamma^{ub}(k) > \Gamma \quad \text{[Saturation]} \\ 0 & \text{if} \quad \gamma^{ub}(k) < 0 \quad \text{[Empty]} \end{cases} \tag{14}$$

Let $\eta(k)$ be the NO_x reduction during a given time step k,

$$\eta(k+1) = u_1(k) - x_1(k+1) \tag{15}$$

Introducing the above definition and the residence time expression (4) into the NO_x process dynamics model (11), we get,

$$\eta(k+1) = \frac{u_1(k)}{F(k)} \tau_0 \; k_{scr}(T(k)) \; \gamma(T(k)) \tag{16}$$

For NO_x process dynamics under catalyst saturation, $\gamma(k) = \Gamma$.

$$\implies \eta_{sat}(k+1) = \frac{u_1(k)}{F(k)} \tau_0 \; k_{scr}(T(k)) \; \Gamma(T(k)) \tag{17}$$

The above equation (17) shows that the tailpipe NO_x concentration becomes independent of urea dosing under catalyst saturation. Polynomial approximations of Arrhenius temperature dependence of the rate constant, k_{scr} and the concentration of viable voids, Γ ([4],[14],[15]), are used to get a linear-in-parameters model for the NO_x process dynamics.

$$k_{scr}(T) = A_{scr} e^{\left(-\frac{E_{scr}}{RT}\right)} \approx \sum k_i T^i \qquad i = 0, 1, \ldots \tag{18}$$

$$\Gamma(T) = S_1 e^{-S_2 T} \approx \sum r_i T^i \qquad i = 0, 1, \ldots \tag{19}$$

The above approximate models for temperature dependence introduce parameter estimates that depend on temperature ranges in the samples. For numerical stability [16], the given temperature range is mapped to $[-1, 1]$ and Chebyshev polynomial bases [17] are used instead of standard polynomials. Incorporating equations (4) and (19) into the NO_x process dynamics model (17),

$$\eta_{sat}(k+1) = \phi_{sat}^T(T(k)) \; \theta_{sat} \tag{20}$$

$$\phi_{sat}(k) = \frac{u_1(k)}{F(k)} \left[2\left(\frac{T-T_0}{T_r}\right)^2 - 1 \quad \frac{T-T_0}{T_r} \quad 1 \right] \tag{21}$$

$$\text{where} \quad T_0 = \frac{T_{max} + T_{min}}{2}, \quad T_r = \frac{T_{max} - T_{min}}{2}$$

It can be observed that in a given time step, the maximum NO_x reduction occurs when the catalyst is saturated. Thus, under the same temperature, flow rate and inlet NO_x concentration, the response of the saturated model (17) forms a tight upper bound to the response of the actual system, which switches between saturation and desaturation based on urea dosing, flow rate and catalyst temperature. This results in the following bounding condition on the response of the saturated system:

$$\eta_{sat}(k) \ge \eta(k) \qquad \forall k \tag{22}$$

This relationship is used for model parameter estimation.

2.1 NO_x sensor cross-sensitivity

Commercial NO_x sensors are cross-sensitive to tailpipe ammonia. Figure 3 shows the effect of cross-sensitivity on commercial NO_x sensors due to tailpipe ammonia. When the tailpipe ammonia is greater than a threshold value, the NO_x sensor output is corrupted, as shown in the plots where the NO_x sensor measurement is significantly higher than the FTIR measurement, which can be considered the true value. The resulting error is modeled as a function of ammonia concentration and a temperature-dependent cross-sensitivity factor $\chi(T)$. This introduces a non-negative directional error $\left(\varepsilon_\chi = \chi\left(T(k)\right) \times [NH_3]^{out} \ge 0\right)$ into the tailpipe NO_x measurement y_1. Accordingly, when the NO_x reduction in a time step is calculated from sensor measurements, $y_1(k) = x_1(k) + \varepsilon_\chi(k)$, the estimated quantity is less than the actual NO_x reduction:

$$\eta_y(k) = u_1(k-1) - y_1(k) = \eta(k) - \varepsilon_\chi(k) \le \eta(k) \tag{23}$$

Thus, the bounding condition on the saturated system's response (22) remains valid when the NO_x reduction during the time step is estimated from the NO_x sensor measurements:

$$\eta_{sat}(k) \ge \eta(k) = \eta_y(k) + \varepsilon_\chi(k) \ge \eta_y(k) \tag{24}$$

Fig. 3 NO_x sensor cross-sensitivity to tailpipe ammonia

3 Maximum Likelihood Estimation of the Model Parameters

The catalyst switches between saturation mode and normal mode based on states that are not completely controllable experimentally. The bounding condition on the saturated system's response (22) can be used for detecting segments of the data where the catalyst's operation is normal or saturated. As the saturated catalyst's response forms a tight upper bound to the actual system's response, the parameters of the system will be the solution to the linear programming problem that minimizes the area under the curve with the lower bound as the response of the actual system:

$$\text{min.} \sum \eta_{sat}(k) \quad \text{s.t} \quad (\eta_{sat}(k) \ge \eta(k)) \quad \forall k \tag{25}$$

The optimization problem (25) is feasible as η_{sat} and $\eta \ge 0$ always. When the NO_x sensor data is used, $\eta(k)$ is replaced with $\eta_y(k)$ as the bounding condition still holds (24). Rewriting the optimization problem using the regression vector (20), we have the linear programming problem for estimating the parameters of the catalyst:

$$\begin{aligned} &\text{minimize} && \mathbf{1}^T \left[\Phi_{sat}\theta_{sat}\right] \\ &\text{subject to} && \Phi_{sat}\theta_{sat} \ge H, \\ &\text{where} && \Phi_{sat} = \left[\phi_{sat}(1), \phi_{sat}(2), \ldots, \phi_{sat}(N-1)\right]^T \\ & && H = \left[\eta(2), \eta(3), \ldots, \eta(N)\right]^T \end{aligned} \tag{26}$$

$\mathbf{1}$ is a column vector of $1's$.

The above estimate of θ_{sat} is the ***maximum likelihood estimate (MLE) under a half-normal distribution [18] for model structure error***, (ε_η), between the saturated system's response, $\eta_{sat}(k)$, and the actual response of the catalyst, $\eta(k)$ (Appendix-A). In the case of NO_x-sensor data with cross-sensitivity, the model structure error includes the cross-sensitivity error ε_χ and is assumed to be a half-normal random variable.

Using asymptotic properties of MLE (Theorem-7.3 in [19]), with sufficiently large data length (N), the parameter estimate $\hat{\theta}_{sat}$ asymptotically follows a normal distribution under some regularity conditions on the PDF $p(\boldsymbol{\varepsilon}_n; \theta_{sat})$,

$$\hat{\theta}_{sat} \sim \mathcal{N}\left(\theta_{sat}, I^{-1}(\theta_{sat})\right) \tag{27}$$

where $I(\theta_{sat})$ is the Fisher information matrix evaluated at the true value of the parameter (θ_{sat}). The regularity conditions require the existence of derivatives of the log-likelihood function and the Fisher information being nonzero. Using the log-likelihood function, $L(\theta_{sat})$, for half-normal distribution (A4), we have,

$$[I(\theta_{sat})]_{ij} = -E\left[\frac{\partial^2 L(\theta_{sat})}{\partial\theta_j \partial\theta_i}\right] = \frac{1}{\sigma^2}\sum_{k=1}^{N}\frac{\partial\varepsilon_\eta(k)}{\partial\theta_i}\frac{\partial\varepsilon_\eta(k)}{\partial\theta_j}$$

$$I(\theta_{sat}) = \frac{1}{\sigma^2}\left[\Phi_{sat}^T \Phi_{sat}\right] \tag{28}$$

The variance in the η_{sat} prediction for the given time-step is obtained by propagating the uncertainty of parameter estimates (27). Using (20), we have,

$$\text{Var}\left[\eta_{sat}(k)\right] = \phi_{sat}(k)^T I^{-1}(\theta_{sat})\phi_{sat}(k) \tag{29}$$

The parameter estimates using FTIR measurements from the test-cell experiments are tabulated in Table 1. The parameters show consistency across repeated RMC (Ramped Mode Cycle) and Hot-FTP (Federal Test Procedure) tests on the same catalyst, which validates the estimation approach. However, the parameters vary significantly between RMC and Hot-FTP tests. This is predominantly due to the differences in the temperature ranges experienced by the catalyst during the tests, which affect the Arrhenius temperature dependence of the rate constant and the concentration of viable voids on the catalyst surface. The temperature range of the data considered during the RMC tests is between $240°C$ and $360°C$, while for the Hot-FTP tests it is between $200°C$ and $300°C$.

Table 1 Parameter Estimates using FTIR Data

Test	$\hat{\theta}_1$	$\hat{\theta}_2$	$\hat{\theta}_3$	σ
DG RMC 1	-0.63	0.55	38.74	1.14
DG RMC 2	-0.82	0.69	38.61	1.08
DG RMC 3	-0.80	1.12	38.78	1.10
Aged RMC	-1.31	1.89	36.94	1.01
DG Hot FTP 1	-2.14	-5.80	39.53	1.35
DG Hot FTP 2	-3.45	-7.99	38.05	1.30
DG Hot FTP 3	-3.37	-6.42	38.88	1.33
Aged Hot FTP	-4.54	-8.01	37.00	1.31

The parameter estimates using NO_x sensor measurements from the test-cell experiments are tabulated in Table 2. The parameter estimates are different from those obtained using FTIR measurements (Table 1) due to the cross-sensitivity of the NO_x sensor to ammonia. However, the parameters show consistency across repeated tests on the same catalyst under similar test conditions. The NO_x sensor cross-sensitivity error $\varepsilon_\chi(T)$ introduces a bias in the parameter estimates, because the component of the error that shares the same temperature dependence as the model structure is absorbed into the parameter estimates. Thus, the linear programming problem estimates the parameters with an added bias term as shown below:

$$\varepsilon_\chi(T) = \phi_{sat}(T(k))^T \theta_{\varepsilon_\chi} + \Delta\varepsilon_\chi \tag{30}$$

$$J(\theta_{sat}) = \sum_{k=0}^{N-1}\left(\phi^T(k)\left(\theta_{sat} + \theta_{\varepsilon_\chi}\right) - \eta(k)\right) \tag{31}$$

Thus, the estimated parameters using NO_x sensor measurements will have a bias dependent on the cross-sensitivity error characteristics of the sensor used. From the experimental results, it is observed that NO_x sensor cross-sensitivity reduces the difference between the estimates of the aging factor (Figures 6 and 7) for aged and degreened catalysts. However, the overall trend of the change in parameter values remains consistent.

Table 2 Parameter Estimates using NO_x Sensor Data

Test	$\hat{\theta}_1$	$\hat{\theta}_2$	$\hat{\theta}_3$	σ
DG RMC 1	-0.70	0.56	38.48	1.10
DG RMC 2	-1.18	0.93	38.43	1.07
DG RMC 3	-0.54	0.62	38.99	1.06
Aged RMC	-1.03	1.26	37.80	1.00
DG Hot FTP 1	-2.30	-6.11	37.08	1.28
DG Hot FTP 2	-2.71	-6.13	36.22	1.27
DG Hot FTP 3	-3.92	-6.17	35.81	1.30
Aged Hot FTP	-6.42	-13.06	34.73	1.34

Furthermore, the parameter estimation method delineated above has a limitation: the data set must include the response of the catalyst under saturation for a finite duration. If this condition is not met, the algorithm estimates the parameters by treating the maximum concentration of adsorbed ammonia achieved in the data as the concentration corresponding to catalyst saturation (Appendix-B).

4 Catalyst Mode Detection

Under catalyst saturation, the NO_x reduction ($\eta_{sat}(k+1)$) is independent of its previous state ($\eta_{sat}(k)$) and initial conditions (20, 21). Rather, it depends entirely on the inputs from the preceding time step, $(u_1(k), F(k), T(k))$. Thus, the measured NO_x reduction at any given time step can be classified as either a saturated or an unsaturated response based solely on these prior inputs. Consequently, by utilizing the estimated parameters, a data point is identified as a saturated-catalyst response if the prediction error for the corresponding inputs (inlet NO_x, u_1, and mass flow rate F) is less than ϵ_{sat}.

$$|\hat{\eta}_{sat}(k) - \eta(k)| \leq \epsilon_{sat} \implies \text{Catalyst Saturation} \tag{32}$$

For the current test-cell and truck data sets, $\epsilon_{sat} = 2.5 \times 10^{-3}\, mol/m^{-3}$. This value is chosen based on the variance of the distribution of the prediction error for the saturated catalyst model.

Fig. 4 Catalyst Mode Detection for DG RMC 1 Data

The catalyst mode detection for the degreened (DG) catalyst under RMC 1 test-cell conditions using FTIR measurements is shown in Figure 4. The figure illustrates the operating points for which the prediction error of the saturated catalyst model satisfies (32), and are therefore classified as saturated, i.e., close to maximum NO_x reduction. The catalyst is found to be close to saturation when the mass flow rate and the urea dosing rates are high, consistent with the physical understanding of the catalyst operation. Furthermore, the flow rate can introduce a virtual ceiling on the maximum achievable NO_x reduction due to reduced residence time in the catalyst, which would be indistinguishable from saturated-catalyst behavior, since the NO_x reduction would be independent of urea dosing.

A fraction of the data from the test-cell experiments under RMC and Hot-FTP test conditions for both degreened and aged catalysts is found to be in saturated mode. The number of data points close to catalyst saturation (N_{sat}) and the total number of samples (N) for different test conditions are listed in Tables 3 and 4. The total

number of data points, N, is the total length of the temporally contiguous segment of the test cell data that is within the chosen temperature operating range for the tests. The operating range for RMC test is between $250°C$ and $350°C$, and for Hot-FTP test is between $200°C$ and $300°C$. Note that the higher number of data points classified as close to catalyst saturation in RMC test conditions, compared to Hot-FTP, results in better parameter estimates and performance of the aging detection test statistic in this case (Table 5).

Table 3 Data Points Close to Catalyst Saturation in Test-Cell FTIR Data

Test	N	N_{sat}	% (N_{sat}/N)
DG RMC 1	2402	490	20.4
DG RMC 2	2402	493	20.5
DG RMC 3	2402	491	20.4
Aged RMC	2402	515	21.4
DG Hot FTP 1	634	186	29.3
DG Hot FTP 2	634	181	28.5
DG Hot FTP 3	638	180	28.2
Aged Hot FTP	636	198	31.1

Table 4 Data Points Close to Catalyst Saturation in Test-Cell NO_x Sensor Data

Test	N	N_{sat}	% (N_{sat}/N)
DG RMC 1	2402	492	20.5
DG RMC 2	2402	494	20.6
DG RMC 3	2402	493	20.5
Aged RMC	2402	515	21.4
DG Hot FTP 1	634	194	30.6
DG Hot FTP 2	634	191	30.1
DG Hot FTP 3	638	198	31.0
Aged Hot FTP	636	220	34.6

5 Model Parameters Capture Catalyst Aging

The parameters of the saturated system are a function of the rate constant for the SCR reaction and the maximum surface concentration of viable voids on the catalyst. The adsorption site concentration is hypothesized to change with catalyst aging, allowing the estimated parameters to capture this degradation. This is demonstrated using the test-cell experiments on a new (degreened) and hydrothermally aged catalyst with the same capacity. The predicted NO_x reduction for an aged catalyst is found to be statistically lower than the predicted NO_x reduction in a degreened catalyst (Figure 5). The dashed line in Figure 5 corresponds to the actual response of the reference system, whose inputs are used to predict the saturated system's response using parameter estimates from the degreened and aged catalyst data under RMC test conditions. Thus, in accordance with the bounding condition (22), the actual response is always less than the predicted saturated system's response. When the η_{sat} response is normalized with respect to the inlet NO_x concentration and flow rate, we obtain a quantity that is only a function of temperature and represents the product of the rate constant and maximum surface coverage. From (20),

Fig. 5 Predicted η_{sat} for Aged and Degreened Catalyst

$$\alpha_{sat} = \frac{F(k)}{u_1(k)}\eta_{sat}(k) = \tau_0 k_{scr}\Gamma = \phi_T^T(k)\theta_{sat} \quad (33)$$

$$\text{where} \quad \phi_T^T(k) = \begin{bmatrix} 2\left(\frac{T-T_0}{T_r}\right)^2 - 1 & \frac{T-T_0}{T_r} & 1 \end{bmatrix} \quad (34)$$

This quantity, α_{sat}, can be estimated and changes with catalyst aging, based on the hypothesis that aging affects the adsorption site concentration. The change in α_{sat} with aging is demonstrated using the test-cell experiments (Figures 6 and 7). The variance is estimated by propagating the variance of the η_{sat} estimate (29). As expected from the cost function for parameter estimation under NO_x sensor cross-sensitivity (31), the parameter estimates have a bias that reduces the difference between the estimates for the degreened and aged catalysts. However, the overall trend of change in parameter values with aging remains consistent across both FTIR and NO_x sensor measurements.

Fig. 6 $\hat{\alpha}_{sat}$ from RMC FTIR Data

Fig. 7 $\hat{\alpha}_{sat}$ from RMC NO_x Sensor Data

Thus, θ_{sat} captures the aging of the catalyst. Hence, the catalyst aging detection problem can be posed as a hypothesis-testing problem, with the null hypothesis ($\mathcal{H}_0$) corresponding to a degreened catalyst and the alternative hypothesis ($\mathcal{H}_1$) corresponding to an aged catalyst. Since the distribution of the error is assumed to be known (half-normal) for both hypotheses and differs by the unknown parameter θ_{sat}, which can be estimated, the following Wald test [20] is proposed to check whether the parameter estimates for a given data set correspond to an aged catalyst or a known degreened catalyst. Decide $\mathcal{H}_1$ if,

$$T_w = \left(\hat{\theta}_{sat} - \theta_{sat}^{dg}\right)^T I\left(\hat{\theta}_{sat}\right)\left(\hat{\theta}_{sat} - \theta_{sat}^{dg}\right) > \beta \quad (35)$$

where θ_{sat}^{dg} is the parameter estimate for the degreened catalyst under the same input conditions, including temperature, flow rate, and urea-dosing rate ranges. The threshold β is chosen based on the experimental results. For the RMC and FTP data sets, the values of the test statistic are tabulated in Table 5. The distribution of the test statistic under aging (non-central χ^2, asymptotically) is different for the RMC and FTP test conditions due to differences in the ranges of inputs (temperature, flow rate, and urea-dosing rate) experienced by the catalyst during the tests. Based on the results, a reasonably discerning threshold that maximizes the probability of detection for both Hot-FTP and RMC test conditions is found to be

Table 5 Wald Test Results on the Test-Cell Data

Test	T_w from FTIR	T_w from NO_x sensor
DG RMC 1	0.26	5.54
DG RMC 2	0.00	0.00
DG RMC 3	6.71	5.46
Aged RMC	47.16	17.68
DG Hot FTP 1	0.49	0.81
DG Hot FTP 2	3.26	1.98
DG Hot FTP 3	0.00	0.00
Aged Hot FTP	4.84	11.44

$$\beta_{hFTP} = 4.00, \qquad \beta_{RMC} = 10.00 \quad (36)$$

Further investigation with more data is needed to determine the probabilities of detection and false alarm for these thresholds. The proposed Wald test detects the aged catalyst for both FTIR and NO_x sensor measurements under both test conditions.

6 Aging Detection from Truck Data

The performance of the proposed aging detector is also demonstrated using the NO_x sensor data from four long-haul trucks. The road data were collected from trucks operating in the United States for two days separated by a few years (Table 6) during which the catalyst had aged based on existing performance measures. The data has measurements of NO_x sensors before and after the catalyst, as well as other required variables including flow rate, temperature and urea injection rate. The data preprocessing includes, 1. removing the rows with missing values, 2. removing the rows whose operating temperature range is beyond the range of $200-300\,^{\circ}C$, 3. interpolating the missing data if the data gaps are smaller than a minute, and finally, 4. smoothing the data using a non-causal Chebyshev filter. The preprocessed data is then partitioned into drive segments with no breaks, which correspond to a continuous driving period from engine start to engine stop. Unlike the test-cell data where the engine drive-cycles are repeatable and produce a significant fraction of samples where the catalyst is close to saturation, the truck data drive segments constitute varied driving conditions with fewer samples $\{\leq 10\%\}$ that are close to catalyst saturation. A necessary condition for the proposed parameter estimation and aging detection framework is that the data has a sufficiently large number of samples close to catalyst saturation such that the asymptotic properties of the MLE and the test statistic hold (37). Hence, the drive segment data are further pruned to only include segments with at least 150 samples (120 in the case of Truck C) close to catalyst saturation.

Table 6 Truck Data Set Years and Mileage Difference

Truck	Earlier Data Year	Latter Data Year	Mileage Difference
Truck A	2015	2017	2.8×10^5
Truck B	2015	2018	11.9×10^5
Truck C	2015	2017	5.9×10^5
Truck D	2015	2016	6.6×10^5

$$N_{sat} \geq N_{sat}^{min} \tag{37}$$

$$\beta_{truck} = 250 \tag{38}$$

The threshold of the Wald test statistic (38) for aging detection in trucks is chosen such that it maximizes the probability of detection while minimizing the probability of false alarm and missed detection for the given limited truck data. The value of the test statistic for the drive segments for each truck is tabulated in Tables 8, 9, 10 and 11. A summary of the aging detection results is tabulated in Table 7. The detector performs reasonably well on three of the four trucks with high probability of detection and low false alarm and missed detection rates. Truck A has relatively lower mileage difference between the earlier and latter data years which may explain the lower detection performance. The performance is limited by the amount of data available with sufficient samples close to catalyst saturation. Further, the lack of ground truth on the actual individual catalyst's aging levels limits the ability to quantify the performance of the detector.

Table 7 Summary of Aging Detection Results on Truck Data

Truck	Total Drive Segments	Detection	False Alarm	Missed Detection
Truck A	8	4	0	4
Truck B	11	9	1	1
Truck C	9	7	1	1
Truck D	8	6	2	0

The threshold $\beta_{truck} = 250$ is significantly larger than the test-cell thresholds ($\beta_{RMC} = 10$, $\beta_{hFTP} = 4$) because the limited sample size causes T_w to diverge from its asymptotic χ^2_ν distribution under $\mathcal{H}_0$. The convergence of the MLE to normality depends on the number of saturated-mode samples [19], and truck data has considerably fewer such samples ($N_{sat} \approx$ 120-150) compared to test-cell experiments ($N_{sat} \approx$ 200-500), resulting in heavier tails in the null distribution of T_w. This effect is compounded by the less reliable estimation of the error variance under road conditions and the unaccounted variance of the reference parameter θ^{dg}_{sat}, which is itself estimated from limited data. Both factors inflate T_w under $\mathcal{H}_0$, necessitating a higher threshold to maintain an acceptable false-alarm rate.

With a larger data set and known degreened paramters, the thresholds can be determined using statistically-optimal selection criteria such as Neyman-Pearson [21] or minimum Bayes cost on the receiver operating characteristic (ROC) curve [22]. This would require multiple truck runs under known catalyst conditions to characterize the null distribution, and runs at known aging levels to estimate the non-centrality parameter under $\mathcal{H}_1$. Nevertheless, the results demonstrate that the proposed saturated-catalyst model (17) coupled with the constrained parameter estimation method (26) and the statistical detector (35) enables a statistically-grounded aging detection framework that performs reasonably well on both test-cell and truck data under sensor sampling limitations and cross-sensitivity.

For on-vehicle implementation, the detector's performance depends on whether the driving profile produces sufficient data close to catalyst saturation within a given driving window. Conditions that combine moderate-to-high exhaust mass flow rates with elevated urea dosing, such as sustained highway cruising, high-load acceleration, or uphill driving within the detector's operational temperature range (200–300 $^{\circ}C$ in the present case, which can be adjusted), are most likely to trigger saturation. The extent of catalyst utilization can be assessed in real time by monitoring readily-available signals such as urea dosing rate, engine rpm, and the estimate of NO_x reduction from sensors. When these indicate sustained low NO_x reduction and low urea dosing which leaves the catalyst unsaturated, the system can determine apriori that the accumulated data are unsuitable for aging detection and defer its decision rather than executing the hypothesis test on insufficient data. The detector therefore avoids producing false alarms under non-catalyst-saturating driving conditions, as the Wald test (35) is simply not invoked. A consequence of this design is that the monitor's in-use performance ratio (IUPMR) is governed by the frequency of saturating driving conditions, since the diagnostic can only complete when the necessary condition (37) is met.

7 Conclusion

The dynamics of NO_x reduction in diesel engine SCR systems under catalyst saturation were used to develop an aging detector based on the hypothesis that the model parameters associated with catalyst storage capacity change as the catalyst ages. A dynamic model for NO_x reduction under saturation was formulated from molar conservation while explicitly accounting for the interplay between the residence time of the reacting flow and the sampling time of the sensors. A maximum-likelihood parameter estimation algorithm was proposed using the bounding condition between the actual system response and the response under catalyst saturation. The algorithm was validated using experimental test-cell data under RMC and Hot-FTP conditions. The NO_x-sensor's ammonia cross-sensitivity, in the absence of a dedicated ammonia sensor, was shown to introduce a bias in the parameter estimates, reducing the separation between the estimates for degreened and aged catalysts. A catalyst mode detection method was utilized to identify data segments operating near saturation. A hypothesis-testing framework based on the Wald test was then developed for catalyst aging detection using the estimated parameters. The detector's performance was demonstrated using test-cell data at different ag-

ing levels and real-world data from four long-haul trucks. The results indicate that the proposed framework can reliably identify catalyst aging even in the presence of sensor cross-sensitivity and varying operating conditions, provided that the data include a sufficient number of samples near the saturation regime, which forms a necessary condition for aging detection.

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Nomenclature

Symbols

$\{\bullet\}$ = Number of moles of $\bullet$. $(moles)$
$[\bullet]$ = Concentration of $\bullet$. (mol/m^3)
A_{scr} = Pre-exponential factor of SCR reaction
E_{scr} = Activation Energy of SCR reaction
F = Mass flow rate of the exhaust gases (g/s)
$I(\theta)$ = Fisher Information Matrix
k_i = Coefficients of polynomial temperature model for k_{scr}
N_{sat} = Number of data points close to catalyst saturation
N = Total number of data points
N_{sat}^{min} = Minimum number of data points close to catalyst saturation for reliable parameter estimation
r_i = Coefficients of polynomial temperature model for Γ
R = Universal gas constant
S_1, S_2 = Coefficients of the temperature model for Γ, $S_1, S_2 \geq 0$
T = Temperature
T_w = Wald test statistic for catalyst aging detection
u_1 = Engine-out (SCR-in) NO_x concentration
V_{scr} = Volume of the catalyst chamber
x_1 = Tailpipe (SCR-out) NO_x concentration
y_1 = Tailpipe (SCR-out) NO_x sensor measurement
α_{sat} = Flow and inlet NO_x normalized NO_x reduction under catalyst saturation
β = Threshold for Wald test statistic for catalyst aging detection
$\gamma(k)$ = Average surface concentration of adsorbed ammonia in the sample [k] to [k+1]
Γ = Surface concentration of the total storage capacity for the catalyst (mol/m^2)
ε_χ = Error in NO_x due to ammonia cross-sensitivity
ε_η = Model structure error between saturated catalyst model and actual system
η_{sat} = NO_x reduction per time-step under catalyst saturation
$\phi_{sat}(T)$ = Regression vector for saturated catalyst model
ρ = Density of the exhaust gases
σ_η = Standard deviation in NO_x
σ = Scale parameter for half-normal distribution
θ_{sat} = Parameter vector for saturated catalyst model
θ_{sat}^{dg} = Parameter estimates for degreened catalyst
χ = Cross-sensitivity factor

Abbreviations

ASC = Ammonia Slip Catalyst
CSTR = Continuously Stirred Tank Reactor
DG = Degreened
FTP = Federal Test Procedure
IUMPR = In-Use Performance Ratio
MLE = Maximum Likelihood Estimate
PDF = Probability Density Function
PFR = Plug Flow Reactor
RMC = Ramped Mode Cycle
ROC = Receiver Operating Characteristic
SCR = Selective Catalytic Reduction

Appendix A: Maximum Likelihood Estimate Derivation

Consider,

$$\varepsilon_\eta(k) = \eta_{sat}(k) - \eta(k) \quad \Longrightarrow \quad \varepsilon_\eta(k) = \left(\phi_{sat}^T \theta_{sat} - \eta(k)\right) \tag{A1}$$

based on the measurement sensors. Assuming ε_η follows half-normal distribution [23] with scale-parameter, σ,

$$\varepsilon_\eta \backsim \text{Half-Normal}(\sigma); \quad p(\varepsilon_\eta;\sigma) = \begin{cases} \frac{\sqrt{2}}{\sigma\sqrt{\pi}} e^{-\frac{\varepsilon_\eta^2}{2\sigma^2}} & \varepsilon_\eta \geq 0 \\ 0 & \varepsilon_\eta < 0 \end{cases} \tag{A2}$$

we have the likelihood and log-likelihood functions given the data and parameter estimate θ_{sat} and σ:

$$l(\theta_{sat}) = p(\boldsymbol{\varepsilon}_\eta;\theta_{sat},\sigma) = \prod_{k=1}^{N} \frac{\sqrt{2}}{\sigma\sqrt{\pi}} e^{-\frac{\varepsilon_\eta^2}{2\sigma^2}} \quad \varepsilon_\eta(k) \geq 0 \tag{A3}$$

$$L(\theta_{sat}) = \ln(l(\theta_{sat})) = \frac{N}{2}\ln\frac{2}{\sigma^2\pi} - \frac{1}{2\sigma^2}\sum_{k=1}^{N}\varepsilon_\eta^2 \quad \varepsilon_\eta(k) \geq 0 \tag{A4}$$

where N is the total sample size. The maximum likelihood estimate for θ_{sat} would be:

$$\begin{aligned} \hat{\theta}_{sat} &= \arg\max_{\theta_{sat}} \left\{ \frac{N}{2}\ln\frac{2}{\sigma^2\pi} - \frac{1}{2\sigma^2}\sum_{k=1}^{N}\varepsilon_\eta^2 \right\} \quad \varepsilon_\eta(k) \geq 0 \\ &= \arg\min_{\theta_{sat}} \left(\sum_{k=1}^{N} \varepsilon_\eta^2 \right) \quad \varepsilon_\eta(k) \geq 0 \\ &= \arg\min_{\theta_{sat}} \left(\sum_{k=1}^{N} \left(\phi_{sat}(k)^T \theta_{sat} - \eta(k) \right)^2 \right) \\ &\quad s.t \quad \phi_{sat}(k)^T \theta_{sat} \geq \eta(k) \quad \forall\, k \in 1,2,\ldots,N \end{aligned} \tag{A5}$$

The quadratic program with a positivity constraint can be relaxed to a linear program as follows. By norm equivalence, the l_2-norm in (A5) is replaced with the l_1-norm, and the nonnegativity constraint renders the absolute value redundant, yielding the linear programming formulation in (A6).

$$\begin{aligned} \hat{\theta}_{sat} &= \arg\min_{\theta_{sat}} \left(\sum_{k=1}^{N} \left(\phi_{sat}(k)^T \theta_{sat} \right) \right) \\ &\quad s.t \quad \phi_{sat}(k)^T \theta_{sat} \geq \eta(k) \quad \forall\, k \in 1,2,\ldots,N \end{aligned} \tag{A6}$$

Appendix B: Limitation: Parameter Estimation Assumes Catalyst Saturation

The catalyst mode detection algorithm presented fails when the catalyst is not saturated for a significant length of time in the data. If the data set does not have any regions of saturation, the approach fits a saturated model as if the maximum achieved catalyst ammonia storage, $(\max(\sigma(k)))$, is the storage capacity of the catalyst (Γ). This is demonstrated using data from high-fidelity simulation model (AVL Cruise [24]) tuned to RMC (Ramped Mode Cycle) test on a test-cell. The RMC test is then simulated with ±20% change in the gain of the urea-dosing controller to change the duration for which the catalyst remains saturated. Figure-8 shows the

Fig. 8 RMC η simulation with ±20% Urea Dosing

Fig. 9 RMC η_{sat} prediction with ±20% Urea Dosing

actual NO_x reduction (η) from the simulation under the three dosing conditions, while Figure-9 shows the corresponding predicted saturated-model response (η_{sat}) using parameters estimated from each case. Figure-8 shows no increase in NO_x reduction when urea dosing is increased by 20%, which indicates that maximum NO_x reduction is already reached at nominal urea dosing. This could be because the catalyst is saturated or because the inlet NO_x concentration is sufficiently low. Conversely, a 20% decrease in urea dosing decreases NO_x reduction. When the −20% urea-dosing data are used to estimate saturated-model parameters, the model predicts a substantially lower NO_x reduction under saturation than for the nominal and +20% dosing cases (Figure-9), the latter two giving essentially identical results. Thus, for the nominal and +20% dosing cases, the catalyst is indeed saturated, and the parameter estimates show the same. However, for the −20% dosing case, the catalyst is not saturated in the data; instead, the parameters fit the maximum surface coverage achieved $\{\theta_{max} = {}^{\gamma_{max}}/_{\Gamma}\}$ in the data, i.e.,

$$\eta_{sat}(k+1) = \frac{u_1(k)}{F(k)}\tau_0\ k_{scr}\Gamma\theta_{max} = \phi_{sat}\theta_{sat} \qquad \text{(B1)}$$

It is assumed that under ordinary drive conditions, the catalyst will reach saturation for a significant length of time, making the proposed parameter estimation approach applicable. In practice, the extent of catalyst utilization over a given data window can be assessed by monitoring readily available signals such as urea dosing rate, engine rpm, and the estimate of NO_x reduction from sensors. to verify whether this assumption holds before executing the detector.

Appendix C: Aging Detector Results on Truck Data Drive Segments

The Wald test statistics, along with the counts and percentages of near-saturation samples, are computed for all qualifying drive segments. Results for Trucks A, B, C and D are provided in Tables 8, 9, 10 and 11, respectively. In each table, entries above and below the horizontal rule correspond to the earlier and latter data years (Table 6). The data demonstrate a strong correlation between successful aging detection and the number of near-saturation samples available in a given segment.

Table 8 Aging Detection Results for Truck A

Drive Segment	T_w	N_{sat}	% $({}^{N_{sat}}/_{N})$
A_15_0	0.00	797	6.70
A_15_1	28.95	305	2.75
A_15_2	58.50	276	5.93
A_15_3	89.56	271	12.77
A_17_0	65.59	414	2.97
A_17_1	127.79	614	6.76
A_17_2	12.13	330	8.44
A_17_3	61.76	336	9.73

Table 9 Aging Detection Results for Truck B

Drive Segment	T_w	N_{sat}	% $({}^{N_{sat}}/_{N})$
B_15_0	672.43	167	1.26
B_15_1	106.66	189	1.81
B_15_3	237.79	159	2.95
B_15_4	0.00	226	6.03
B_15_6	84.73	179	5.75
B_18_0	543.55	231	2.10
B_18_1	1050.24	308	3.00
B_18_4	22.48	195	6.14
B_18_6	7150.19	378	12.73
B_18_7	760.89	224	8.14
B_18_8	342.67	246	9.04

Table 10 Aging Detection Results for Truck C

Drive Segment	T_w	N_{sat}	% $({}^{N_{sat}}/_{N})$
C_15_1	0.000	186	1.30
C_15_4	113.86	143	3.84
C_15_5	4340.08	124	4.16
C_17_0	700.88	186	1.24
C_17_2	3993.85	446	7.23
C_17_3	1693.49	199	4.00
C_17_4	186.64	120	2.81
C_17_7	1878.52	129	4.73
C_17_9	7500.10	218	11.63

Table 11 Aging Detection Results for Truck D

Drive Segment	T_w	N_{sat}	% $({}^{N_{sat}}/_{N})$
D_15_0	0.00	214	1.29
D_15_2	98.99	160	2.95
D_15_4	1546.55	213	4.90
D_15_6	935.05	152	8.86
D_16_1	423.20	246	3.67
D_16_2	1022.06	199	5.22
D_16_3	697.78	170	5.36
D_16_5	313.62	200	9.31

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